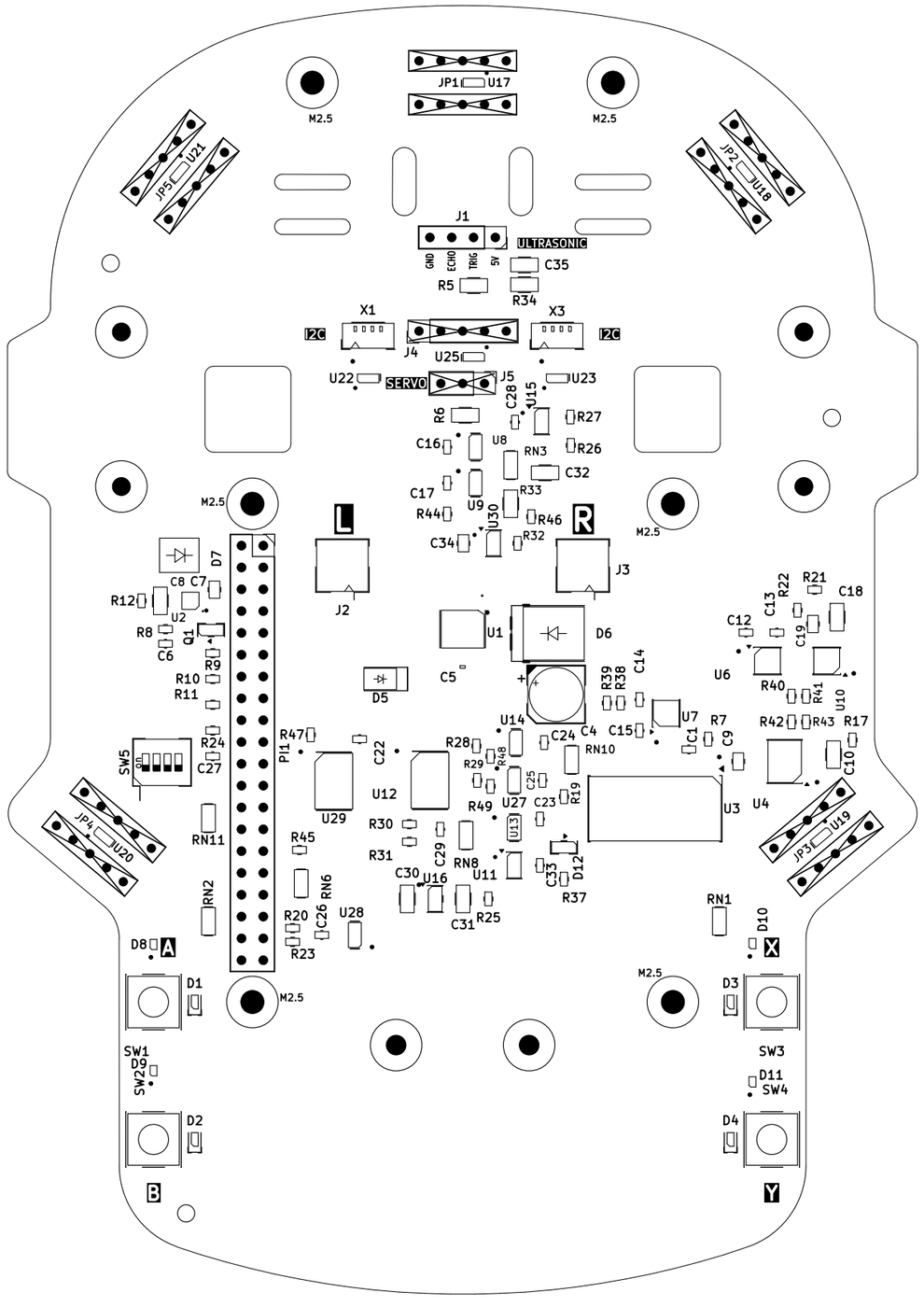


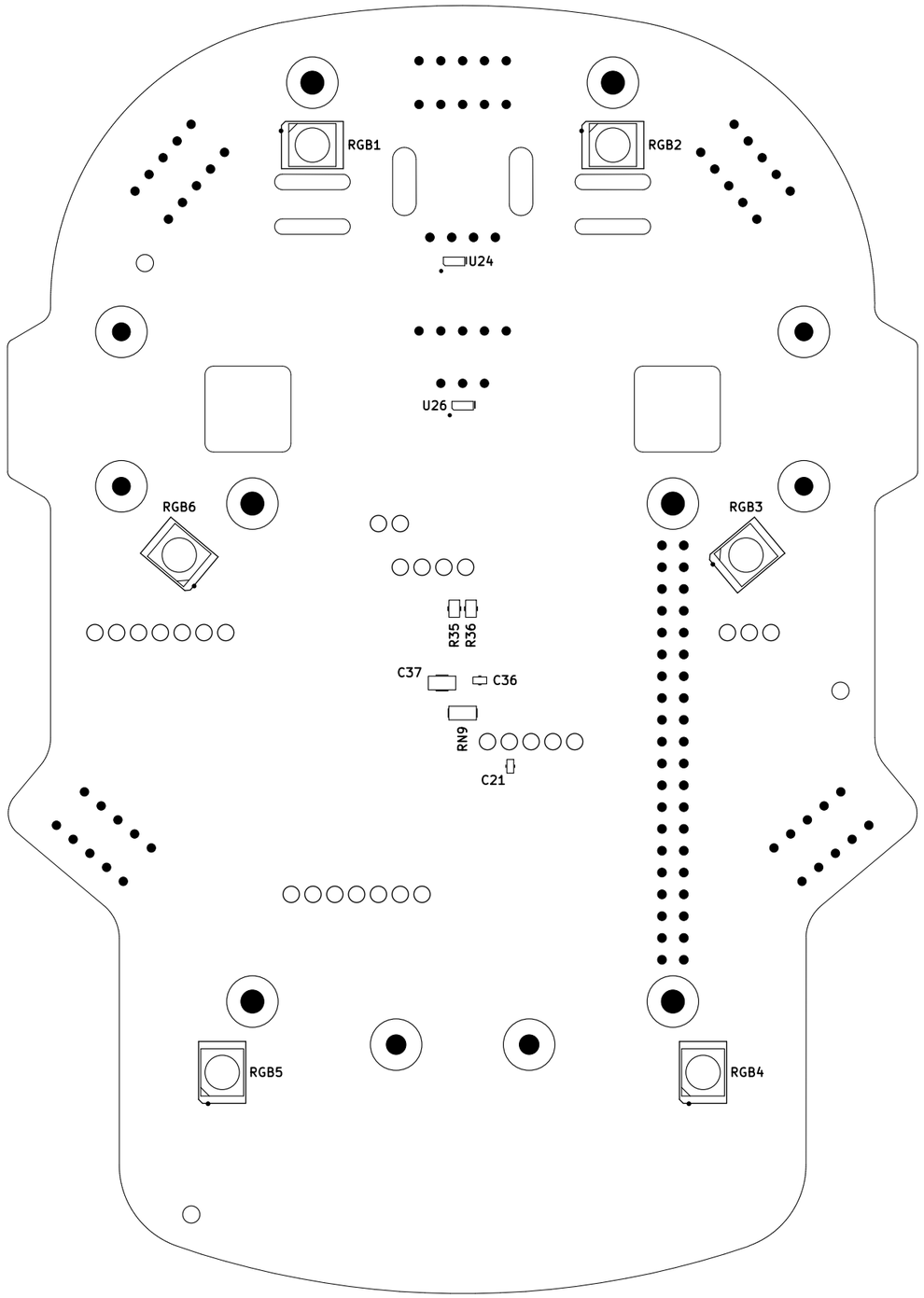
TOP SIDE

SCALE 3:2

NOTES

1. Top side, viewed from above. Scale 3:2.
2. Reflow the bottom side (sheet 2) first, then this side.
3. Place the SMD parts from Keel_R3_CPL.csv. Part numbers: Keel_R3_BOM.csv.
4. PI1 and J1 are through-hole parts. Hand-solder them after reflow; no wave soldering. Seat them flat. PI1 pins 2 and 4 carry the board current and need full hole fill.
5. Crossed-out positions are not fitted: J4, J5 and JP1–JP5. Leave their holes open. U17–U21 sit inside the JP outlines and are fitted.
6. Pin 1 is at the silk dot or the chamfered body corner. C4: + marks the positive terminal.
7. D1–D4: the body mark is the anode and goes to pad 2, away from the silk bracket.
8. U17–U26: pin 1 is the end with the silk dot. The short bar at the other end is not a pin-1 mark.
9. D6 is the unidirectional SM15T6V8A; its cathode is at the bar.
10. U12 is fitted blank. It is programmed after assembly.
11. SW5: all four poles OFF (RUN). Keep the tape seal on.
12. Plain small circles are the fiducials FID1–FID3.

Raspberry Pi 4 robot baseboard Scale 3:2 CERN-OHL-P-2.0 **BisocM**	
Sheet: Top side File: Keel.kicad_pcb	
Title: Keel R3 assembly drawing – top side	
Size: A3	Date: 2026-11 **Rev: R3**
KiCad E.D.A. 10.0.6	Id: 1/2



BOTTOM SIDE — VIEWED FROM BELOW

SCALE 3:2

NOTES

1. Bottom side, viewed from below (mirrored). Scale 3:2.
2. Reflow this side first. 14 parts: RGB1–RGB6, U24, U26, R35, R36, C21, C36, C37 and RN9.
3. Place the SMD parts from Keel_R3_CPL.csv. Bottom-side coordinates and angles are not mirrored: orient the parts by this sheet.
4. RGB1–RGB6: pin 1 is at the silk dot and the chamfered body corner.
5. U24, U26: pin 1 is the end with the silk dot.
6. Rows of circles are the bare test pads TP1–TP47: no paste, no parts.
7. Plain small circles are the fiducials FID4–FID6.
8. The two pin rows of PI1 and the four pins of J1 belong to the through-hole parts on sheet 1. The other hole rows are the unfitted positions J4, J5 and JP1–JP5.

Raspberry Pi 4 robot baseboard
Scale 3:2

CERN-OHL-P-2.0
BisocM

Sheet: Bottom side, viewed from below
File: Keel.kicad_pcb

Title: Keel R3 assembly drawing — bottom side

Size: A3	Date: 2026-11	**Rev: R3**
KiCad E.D.A. 10.0.6		Id: 2/2